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Initial Post

by [Ali Alhammadi](#) - Friday, 8 August 2025, 12:08 PM

According to Metcalf (2024), Industry 5.0 builds upon the automation-oriented emphasis of Industry 4.0 by creating a balance between new technology and resilience, ethics, and sustainability. This shift has serious ramifications in the aviation field where safety, reliability, and client confidence are of the essence. The 2024 CrowdStrike IT outage, where a security software update went wrong and led to massive system failures across industries, resulting in grounded operations, could be a recent example of where Industry 5.0 principles must be employed. One of the most impacted was Delta Air Lines, as more than 7,000 flights were cancelled within five days, disrupting more than 1.3 million passengers (Alzaabi, 2024). The U.S. Department of Transportation opened a probe into airline readiness, crisis response, and civil suits, such as a \$500 million case, citing negligence in the system risk mitigation (Alojaiman, 2023).

Effects were not only immediate losses in operations. This caused inconvenience to customers, a dent in reputational trust in the airline, and an impact on the economy, spreading to related industries like hospitality and tourism. The financial loss due to the outage was estimated globally in tens of billions of U.S. dollars (Moenck et al., 2025). The incident showed that a single point of failure can become a cascade across the interconnected networks in well-centralized, homogeneous IT environments. Regarding Industry 5.0, this failure reaffirms the need to design information systems that exhibit redundancy, decentralization, and human supervision. If Delta and its technology partners had diversified their infrastructures and had more robust human-in-the-loop protocols, the extent of upheaval might have been mitigated. The challenges directly address the industry 5.0 focus on socio-technical resilience and ethical collaboration between technology providers and operators (Abdel-Basset et al., 2025; Alojaiman, 2023). Since aviation is increasingly shifting to more integrated digital systems, it will be imperative to implement these principles to create operational resiliencies and public confidence in the wake of unexpected technological breakdowns.

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Peer response

by [Yousif Ali Karam Yousif Almaazmi](#) - Sunday, 10 August 2025, 5:49 PM

Your analysis clearly shows how Industry 5.0 principles relate to the vulnerabilities highlighted by the hypothetical 2024 CrowdStrike IT outage. Your example of the Delta Air Lines case demonstrates how cascading IT systems failure can be seen in centralized and homogeneous IT systems with a single point of failure. I especially appreciate your connecting this with socio-technical resilience, redundancy, and human-in-the-loop protocols, which fall at the center of Industry 5.0 thinking (Abuhasel, 2023). This detailed focus points out the losses within operations. It demonstrates the broader reputational, legal, and economic consequences of such misfortunes, demonstrating that such failures extend beyond mere operational disturbance.

It is right on about decentralization and diversified infrastructure. Such safeguards are essential as the aviation industry continues to rely highly on digital systems that work closely together. I also like that you have considered the immediate passenger and wider ripple effects into adjacent industries such as hospitality, tourism, and cargo-based logistics (Alzaabi, 2024). Such indirect effects are frequently disregarded, and one should pay close attention to them since they increase the significance of implementing resilient systems that can continue to work to some degree even in the case of vital failures.

To further ground your post, you might add more detailed resilience measures, like multiple backup control centers, real-time anomaly detection, multi-vendor infrastructural diversity, and compulsory system tests during a simulated crisis. Investigating ways that technology suppliers and users might potentially create unified crisis management systems, including preset communication channels, would also be beneficial to adding effectiveness to your claim on ethical collaboration (Atif & Qureshi, 2024). Your example has argued that manufacturing 5.0 is not merely about introducing new technologies but is concerned with designing for stability, equitability, and trustworthiness in the first place so that all humans and automated systems can adapt to sudden events.

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Peer response

by [Abdulla Husain Salem Hadna Almessabi](#) - Monday, 11 August 2025, 10:06 AM

Dear Ali Alhammadi

You point to the burning issue of even the principles of Industry 5.0, with the realities of the work in aviation, especially after the CrowdStrike experience in 2024. The failure shows the risks posed by centralised IT systems in terms of efficiency, in that they may become susceptible to cascading failures that affect the operations, which in the example goes further to affect trust of its customers and the general sector of the economy (Khern-am-nuai, 2024).

The case shows the ripple effect of one point of failure, which highlights the significance of having effective risk management structures, an area that is outlined in the transition to Industry 5.0 (Mugu et al., 2024). As you propose, the scale of the crisis could have been reduced using diversified infrastructures and incorporating human-in-the-loop protocols.

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As Industry 5.0 pursues the building of socio-technical resilience, it would appear imperative to move toward more adaptive, ethical partnerships between technology vendors and the organisations that use them (Costa et al., 2025). Greater resiliency in the tech ecosystem of the airline industry needs to be focused on to ensure the occurrence of such incidents is avoided again and confidence is built back amongst consumers, particularly during such disruptions of critical nature (Ivashchuk and Ostroumov, 2024). The future-proofing of the technological infrastructure of the aviation industry depends on the integration of the principles of Industry 5.0 (Yimga, 2025).

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Peer Response

by [Saleh Almarzooqi](#) - Sunday, 17 August 2025, 8:50 PM

The CrowdStrike IT outage on Delta Air Lines demonstrates how the Industry 5.0 concepts, especially the prioritisation of resilience, decentralisation, and human control, can be of utmost importance (Amorim, Fernandes, and Filipe, 2025). The outage is a severe reminder of the weaknesses in spite of highly interconnected and centralised IT systems. The impacts of a point of failure can be so destructive not only to the concerned airline but also to the whole ecosystem of the stakeholders, such as the passengers, hospitality providers, and tourism industries (Mugu et al., 2024).

This is something that, as you rightfully indicated, could have been avoided by having redundancies in systems and a greater form of decentralisation of infrastructures so that the outage did not get out of hand. An extra step, embarked with a human in the loop, or motivated to make certain that operators monitor and make decisions in the system, would have added another degree of resilience in the system.

The idea of human-centricity in Industry 5.0 implies that the usage of AI and automation is not supposed to negate the use of human judgment but supplement it. Having human supervision of important processes will help the industry mitigate the consequences of possible failures and not become as chaotic as in this case (Nasir, 2025). The crisis also highlights the ethical aspect of the technology providers who are expected to incorporate robust and accountable ideas into their systems.

The financial losses and reputational ruin that you talked about point out an essential aspect: companies must implement a strategy of resilience and risk mitigation in the era of sophisticated technology. Industry 5.0 does not necessarily signify new technology adoption, but engages in keeping society in a good state by utilising new technologies, while keeping this with a mind of society.

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Nasir, M., 2025. Impact of CrowdStrike Outage on Delta Air Lines.

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